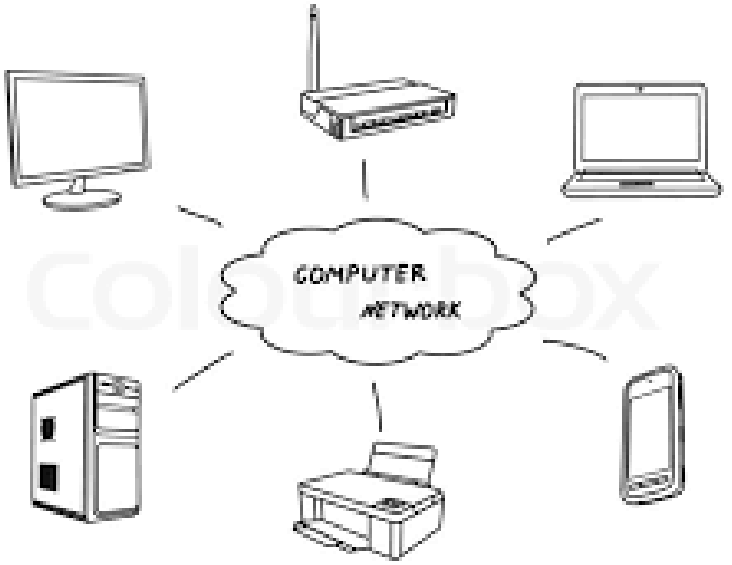
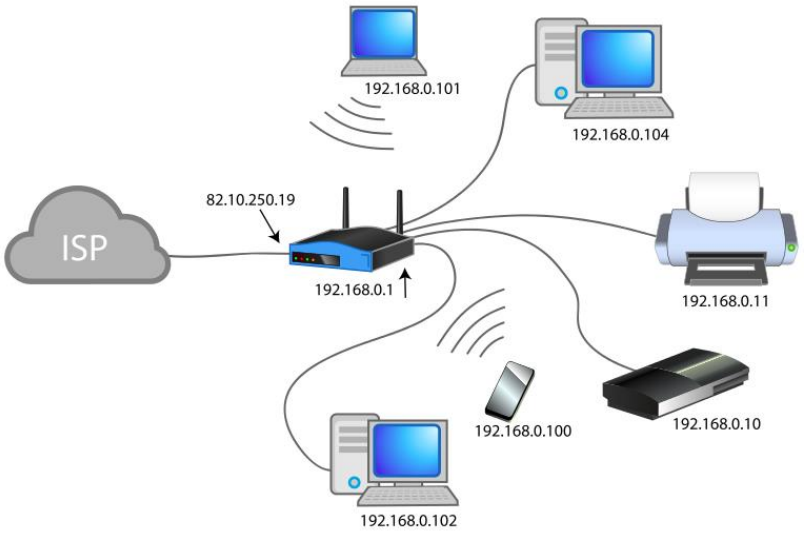


Orientation Programme
On
Computer Networks Lab
(Needs of Day – to – Day Networking)

Networking Components

What is Computer Networks?



Computing devices connected through media like Co-axial cable, Twisted pair or Radio waves and able to communicate each other

Networking Components (Cntd.,)

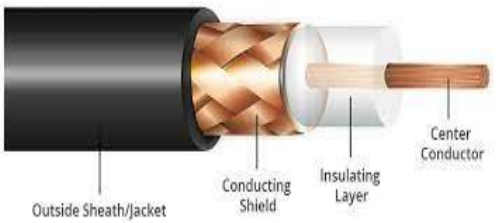
How media is interfaced to Computer?



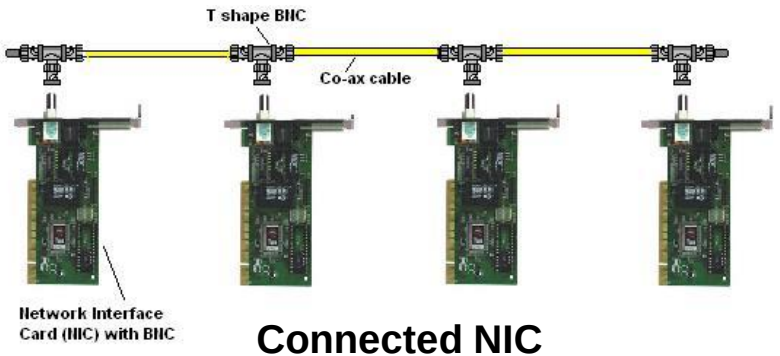
NIC with BNC on Computer side



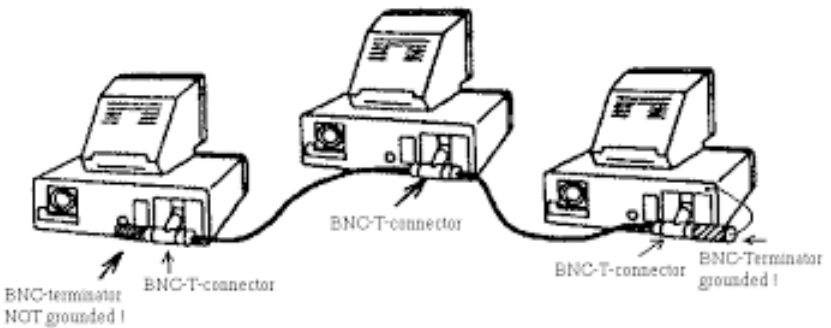
BNC connector on cable side



Co-axial cable

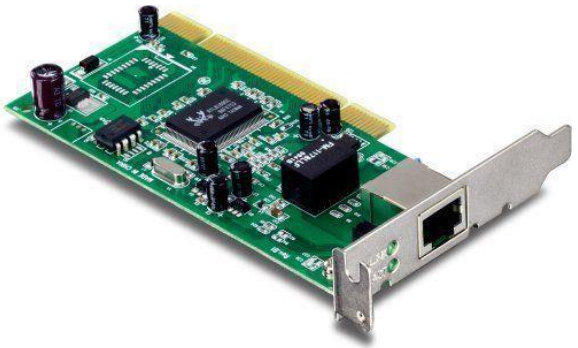


Connected NIC



Computer network using BNC (Bayonet Neill–Concelman)

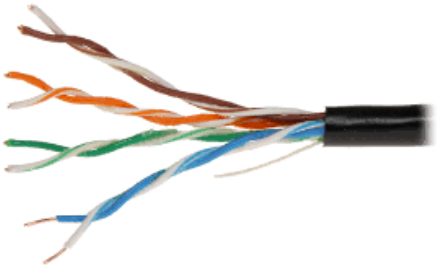
Networking Components (Cntd.,)



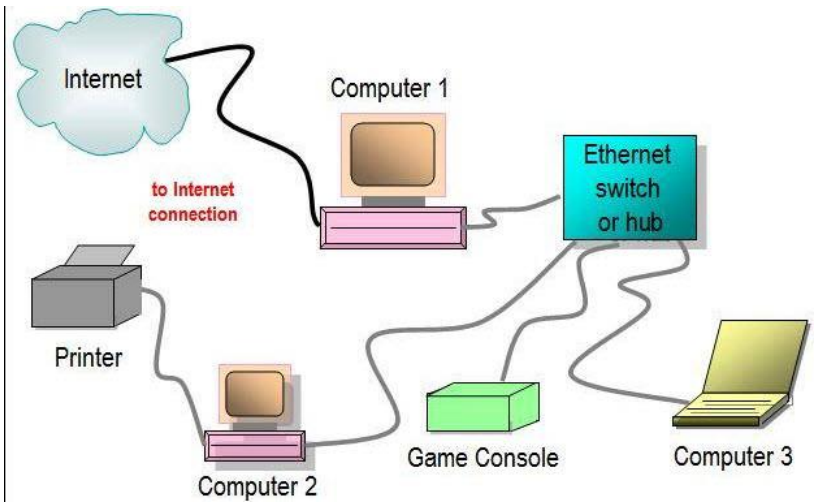
NIC with RJ-45 on Computer side



RJ-45 Connector on cable side



Twisted pair cable



Computer network using RJ-45

Networking Components (Cntd.,)



NIC with Wi-Fi on Computer side



USB – Wi-Fi Module

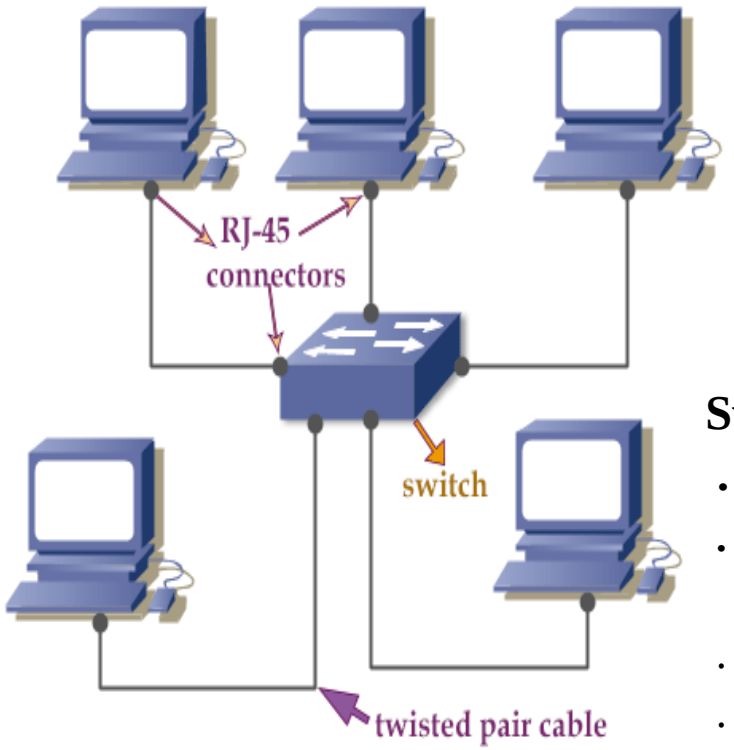
Frequency Band:2.4-2.4835 GHz
Condition – Indoor
- Low power wireless equipments
Application - Wi-Fi

ISM freuency range used



Wireless Local Area Network

Ethernet Star Topology (802.3)



NIC Card

- 48-bit MAC/Hardware address
- Data-link layer(MAC layer) protocol use this address to identify the system on the network
- Two types of NIC - wireless NIC and wired NIC.

Switch

- Connects multiple devices together
- Allow devices to share and transfer data using MAC address
- An Ethernet switch works at the data link level
- Two types switches Unmanaged and Managed

Ethernet Star Topology (Contd..)



Unmanaged Swtch

. Plug and Play device

. Requires no special installation

. Auto-negotiation between Ethernet devices

. Switch will automatically determine the best data rate to use

. Allows to create subnet



Managed Swtch

. Gives control over the traffic on network

. Allows admin to customize & configure each Ethernet port to get maximum efficiency over data transfers on the network

. Managed switches allows to create VLANs

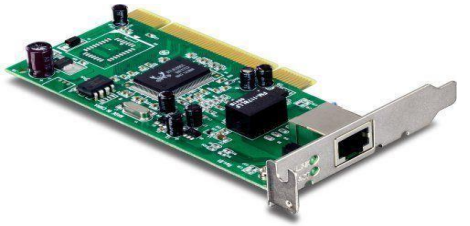
. Higher in cost

. Allows port mirroring

. Managed switches have a remotely accessible console to configure from different physical locations.

. Uses SNMP (Simple Network Management Protocol) to monitors all devices connected

Numbers in Computer Networks



MAC/Physical/Hardware address:

- 48 bit Hex-decimal number embedded into NIC during manufacturing
- MAC Address is worldwide unique
- MAC address is used by the Media Access Control (MAC) sub layer of the Data-Link Layer.
- MAC Address is a 12-digit hexadecimal number separated by semicolon, hyphen or dot
- MAC address is permanent which cannot be changed
- Actual communication happens on the network based on MAC address

Numbers in Computer Networks (Contd.,)

MAC/Physical/Hardware address:

- The First 6-digits of MAC Address identifies the manufacturer, called OUI (**Organizational Unique Identifier**), IEEE Registration Authority Committee assigns these MAC prefixes
- Example
    - CC:46:D6 - Cisco
    - 3C:5A:B4 - Google, Inc.
    - 3C:D9:2B - Hewlett Packard
    - 00:9A:CD - HUAWEI TECHNOLOGIES CO.,LTD
    - 34-17-EB - Dell Inc.
- Example of MAC 34-17-EB-54-C8-EB
- ipconfig –all and ifconfig commands are used to get Hardware address of NIC

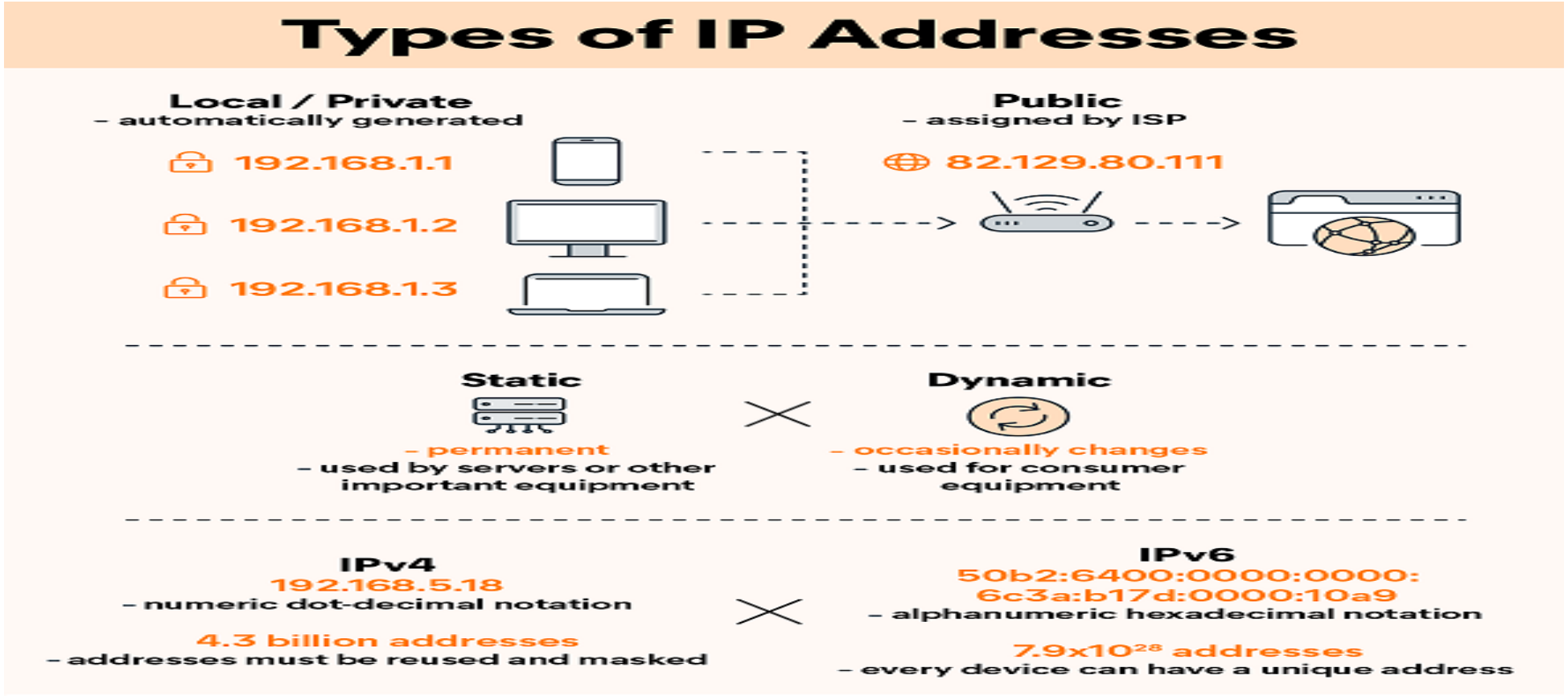
Numbers in Computer Networks (Contd.,)

IP (Internet Protocol) Address:

- An IP address is a unique address that identifies a device on the internet or LAN
- IP is 32 bit address represented by 4 octets separated by dot
- Example 192.168.16.34 11000000.10101000.00010000.00100010
- IP address is used in Network Layer/ Internet Layer
- IP address can be changed
- Allocated by the Internet Assigned Numbers Authority (IANA), a division of the Internet Corporation for Assigned Names and Numbers (ICANN).

Network Id	Host Id

Numbers in Computer Networks (Contd.,)



Numbers in Computer Networks (Contd.,)

Public IP:

- IP that can accessed directly over the internet and assigned by ISP
- Public IP can not be reused by other devices
- Assigned and controlled by your Internet Service Provider
- Any IP not included in the reserved private IP address range

Private IP:

- A private IP address is the address that identifies the device on the Local network and is assigned by Admin or your network router
- Private IP can be reused by the devices in the other network

  Class A: 10.0.0.0 — 10.255.255.255
  Class B: 172.16.0.0 — 172.31.255.255
  Class C: 192.168.0.0 — 192.168.255.255 are the reserved private IPs

Basic Networking Commands

hostname : this command will display the name of your computer
c:\user\admin>*hostname*

ipconfig : this command will display the details of all the adaptors present in the computer

c:\user\admin>*ipconfig* display the basic TCP/IP configuration for all adapters, type

c:\user\admin>*ipconfig /all* display the full TCP/IP configuration for all adapters, type

ping: this command line utility to check the connectivity between the source and destination host
It uses Echo request and Reply messages to check the connectivity.

ping -n 5 -l 1500 www.google.com
sends 5 request packets with the packet length of 1500 bytes instead of 32 bytes

ping 127.0.0.1
It will test the NIC of local host

Basic Networking Commands

Address Resolution Protocol (ARP)is the protocol that bridges Layer 2 and Layer 3 of the OSI model,

ARP resolves IP addresses into MAC addresses on Ethernet networks

Once a MAC and IP address pair is discovered , it's kept in the ARP table for a specified period of time.

c:\user\admin>*arp -a* will also show the ARP table

c:\user\admin>*arp -d* will delete an entry in ARP table

c:\user\admin>*arp –s* <ip> <MAC> adds pair to the table

Basic Networking Commands (Contd.,)

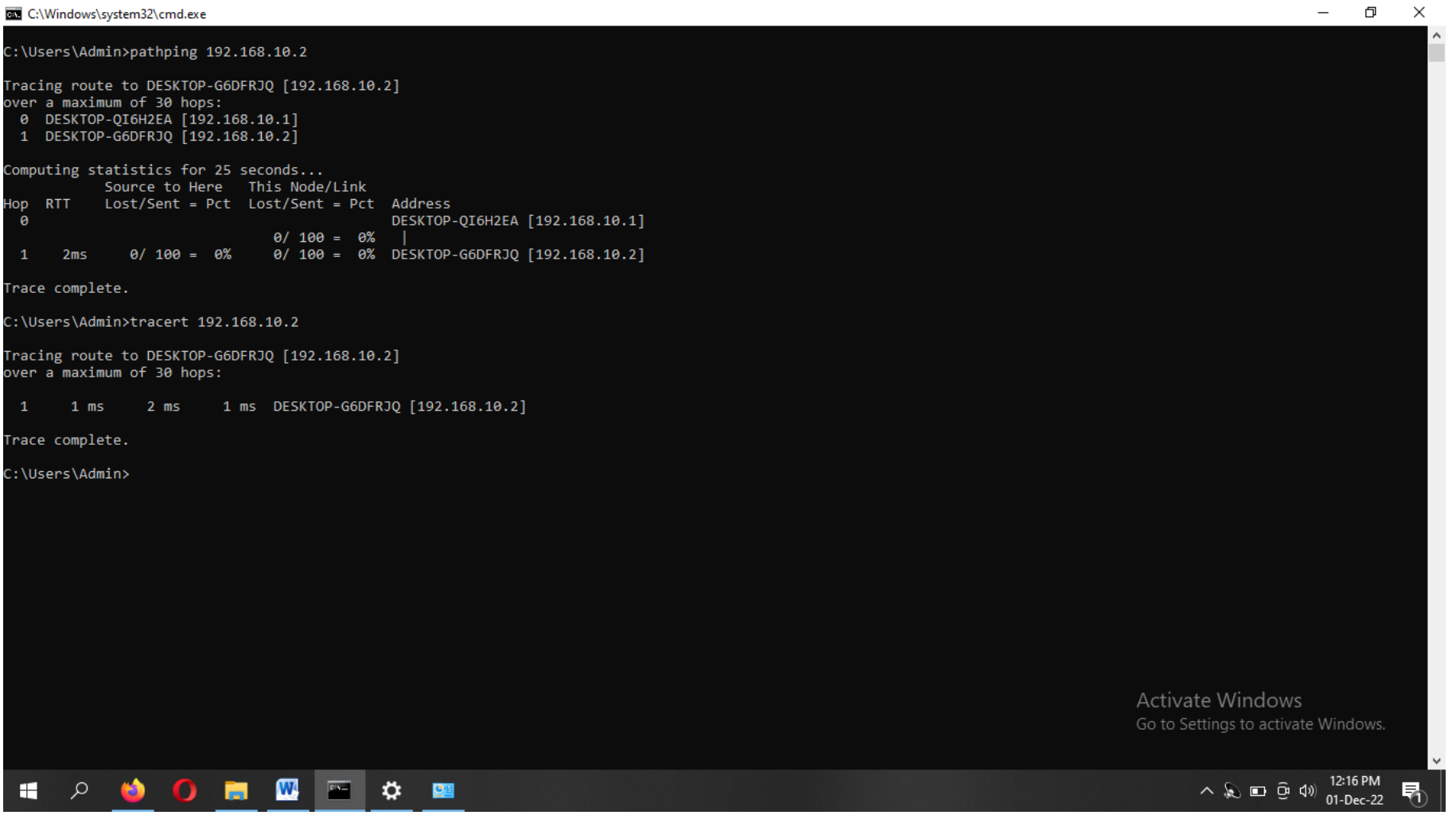
pathping: Sends echo Request messages to router between a source and destination, over a period of time, and then computes results based on the packets returned from each router.

- It is a tool used to troubleshoot network issues.
- pathping is a command that provides information about network latency and network loss at intermediate hops between the source and the destination devices.
- **c:\user\admin>*pathping rvce.edu.in***

tracert helps to find the actual path from the source to the destination device

- It prints out an ordered list of the intermediate routers that return ICMP
- ***C:>tracert 192.168.1.1***
- ***C:>tracert -h 3 lifewire.com > z:\tracertresults.txt***

Basic Networking Commands *(Contd.,)*



Basic Networking Commands (Contd.,)

netstat provides statistics about all active connections in the current network

Generates displays network status and protocol statistics.

C:>netstat : Standard listing of all active connections

C:>netstat –a : Displays all active ports

C:>netstat –e : Shows statistics about your network connection (received and sent data packets, etc.)

C:>netstat –n : Numerical display of addresses and port numbers

```
Proto  Local Address          Foreign Address        State
TCP    127.0.0.1:50473        DESKTOP-QI6H2EA:50474  ESTABLISHED
TCP    127.0.0.1:50474        DESKTOP-QI6H2EA:50473  ESTABLISHED
TCP    127.0.0.1:61138        DESKTOP-QI6H2EA:61139  ESTABLISHED
TCP    127.0.0.1:61139        DESKTOP-QI6H2EA:61138  ESTABLISHED
TCP    127.0.0.1:61140        DESKTOP-QI6H2EA:61141  ESTABLISHED
TCP    127.0.0.1:61141        DESKTOP-QI6H2EA:61140  ESTABLISHED
TCP    172.16.64.173:52685    ec2-34-199-53-229:https  ESTABLISHED
TCP    172.16.64.173:52692    65:https               ESTABLISHED
TCP    172.16.64.173:52693    139:https              ESTABLISHED
TCP    172.16.64.173:52694    137.184.239.188:https  ESTABLISHED
```

Note: for all option with netstat

Basic Networking Commands (Contd.,)

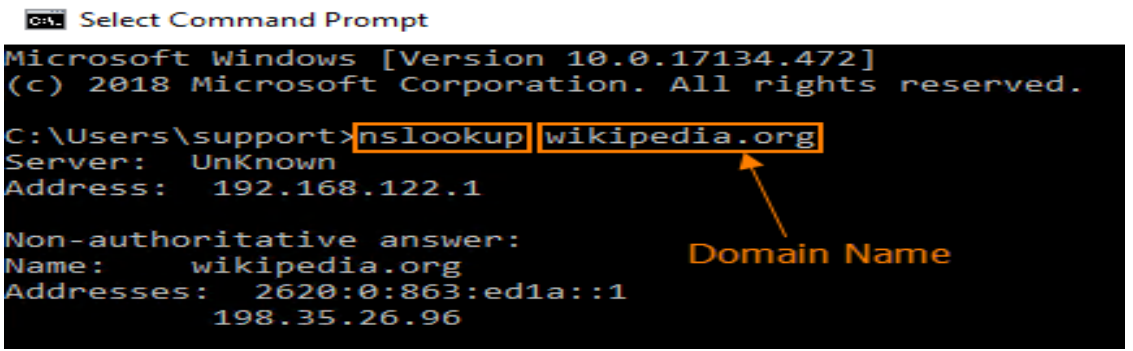
nslookup: This tool can be used to check DNS records propagation

Displays local DNS server and its IP address.

```
Microsoft Windows [Version 10.0.18363.1679]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\support>nslookup
Default Server:  UnKnown
Address:  192.168.122.1
```

Displays the information of the domain we specify.

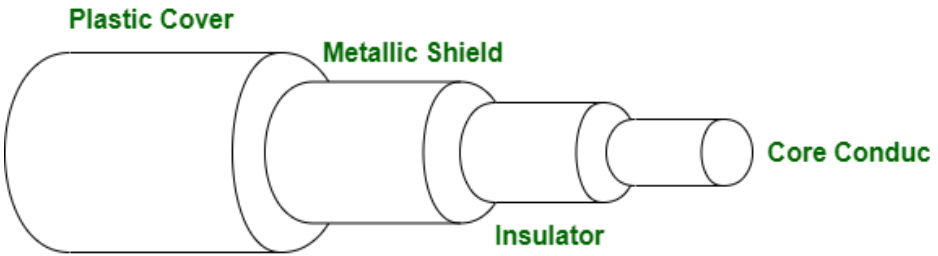


Display the name servers for the domain you specified.

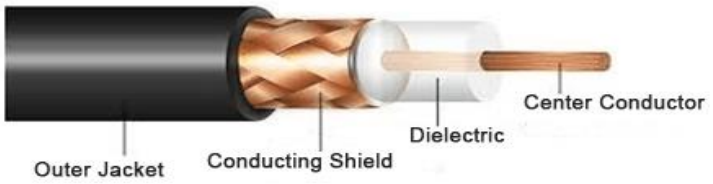
```
c:\>nslookup -type=ns wikipedia.org
Server:  UnKnown
Address:  10.200.18.22

Non-authoritative answer:
wikipedia.org    nameserver = ns0.wikimedia.org
wikipedia.org    nameserver = ns1.wikimedia.org
wikipedia.org    nameserver = ns2.wikimedia.org
```

Types of Cables

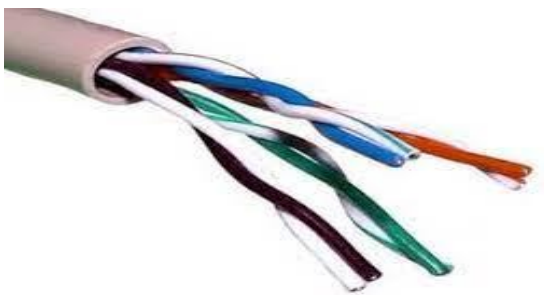


Coaxial Cable





Twisted pair DNA structure

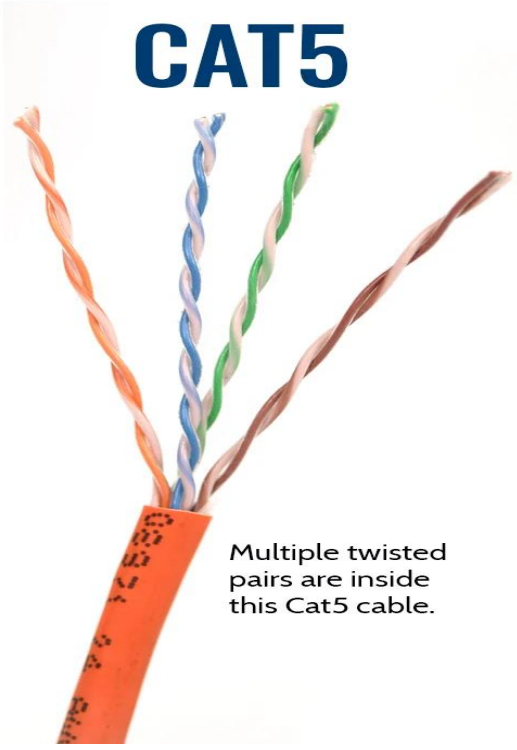


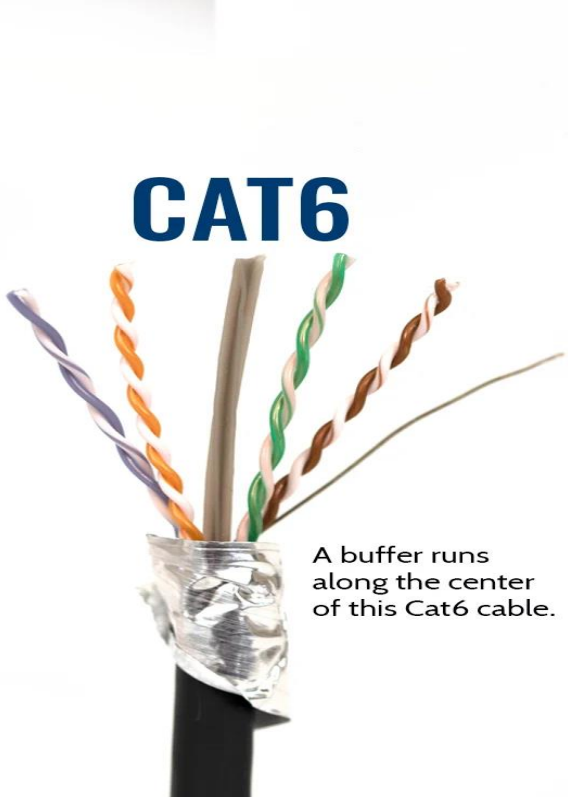
Unshielded Twisted pair



Shielded Twisted pair

Types of Cables







Types of Cables

UTP Cat5, has a transfer rate of 100mbit/s and a bandwidth of 100Mhz

UTP Cat5e, has a transfer rate of 1.000mbit/s and a bandwidth of 100Mhz

UTP Cat6, has a transfer rate of 1.000mbit/s and a bandwidth of 250Mhz

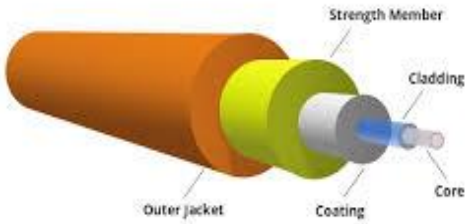
UTP Cat6a, has a transfer rate of 10.000mbit/s and a bandwidth of 500Mhz

UTP Cat7, has a transfer rate of 10.000mbit/s and a bandwidth of 1.000Mhz

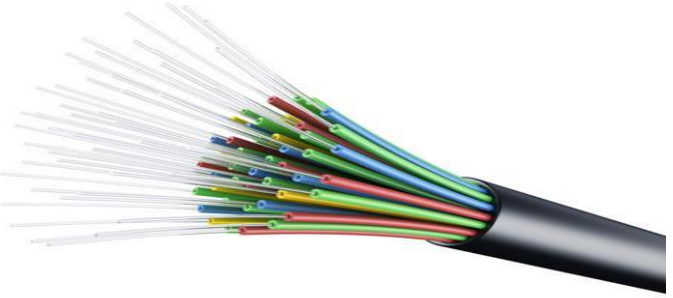
UTP Cat7a, has a transfer rate of 10.000mbit/s and a bandwidth of 1.200Mhz

UTP Cat8 has a transfer rate of 10.000mbit/s and a bandwidth of 2.000Mhz

Types of Cables



Fiber Optic cable structure



Fiber Optic cable





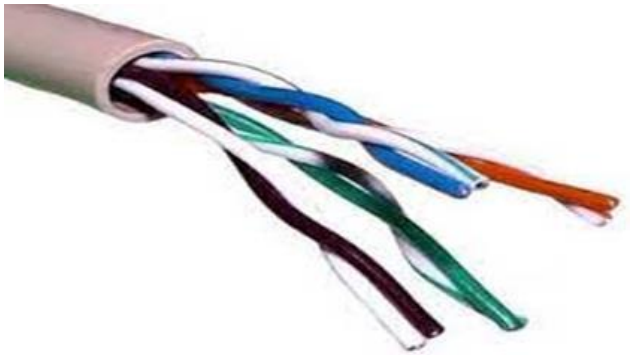
Throughput –data rates can reach up to 100 gigabits per second per channel.

Connectors – With fiber cabling, you can use any of 10 different types of connectors

Size and scalability – Depending on the type of fiber optic cable used, segment lengths vary from 150 to 40,000 meters.

Noise immunity –, it is unaffected by EMI.

Crimping CAT cable

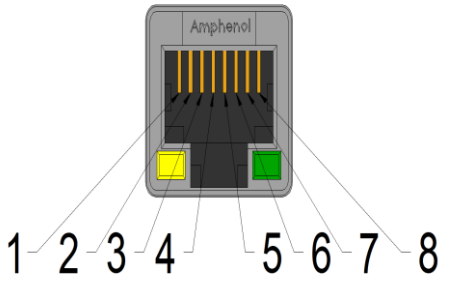


Cat Cable



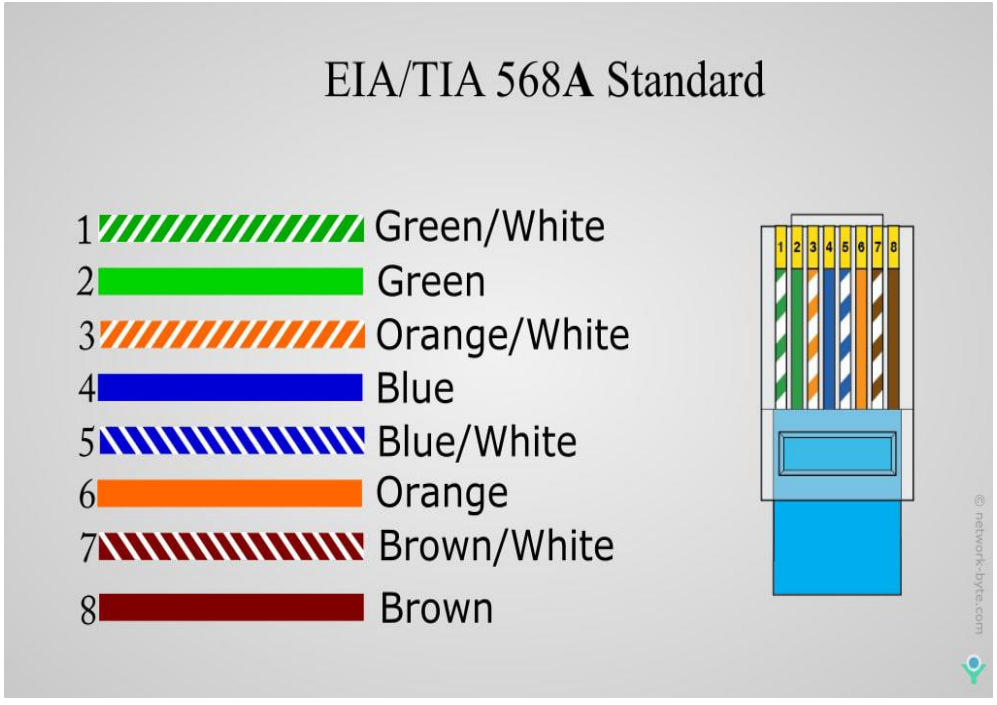
Registered Jack-45 Connector

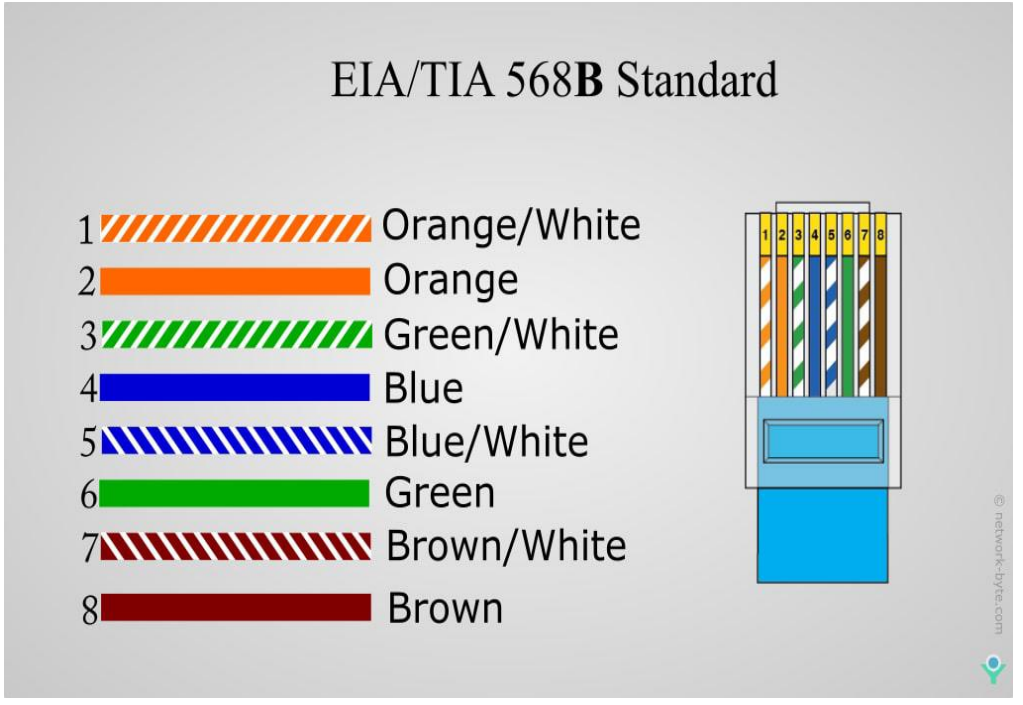
- Each colored wire is twisted with white wire with a strip of same color on it
- There are 4 twisted pairs with four different colored wires
- Only Two pairs are used with cable numbers 1, 2, 3, and 6 for Tx and Rx signals



Crimping CAT cable

As per EIA(Electronic Industries association) and TIA(Telecommunication Industry Association) there are two standard EIA/TIA-568-A and EIA/TIA-568-B





Crimping CAT cable

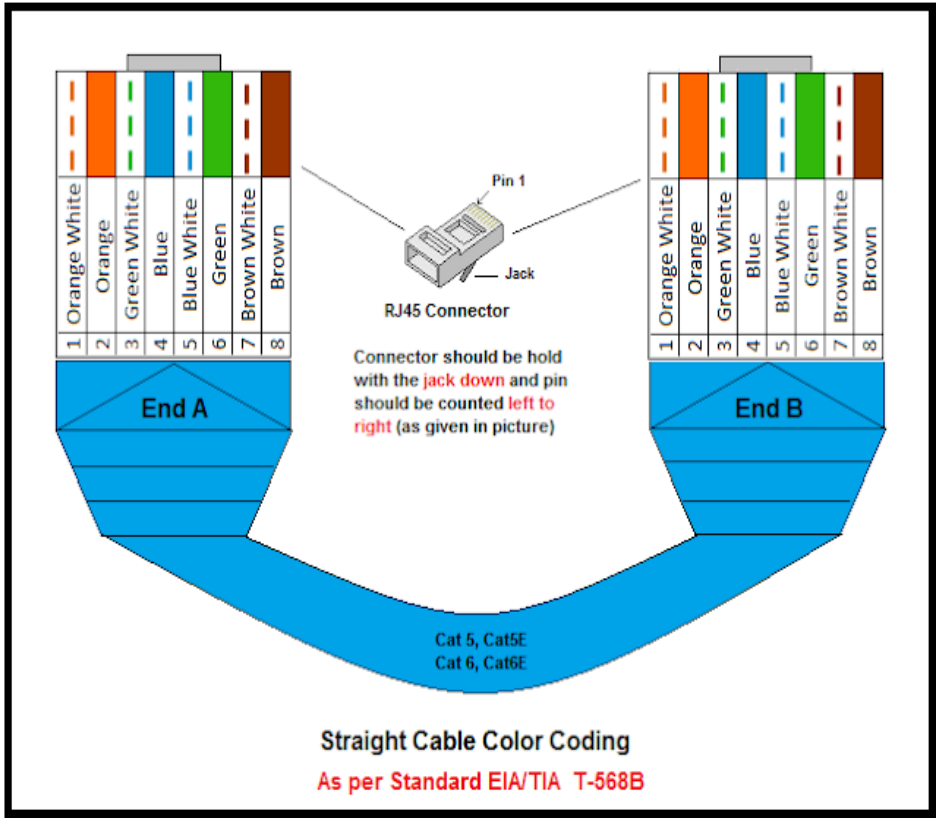
Straight Cable

Used to connect devices with different function.

- Connecting a router to a hub or switch.
- Connecting a server to a hub or switch.
- Connecting workstations to a hub or

Left	1	Tx+			Rx+	1	Left
	2	Tx-			Rx-	2	
	3	Rx+			Tx+	3	
	4					4	
	5					5	
	6	Rx-			Tx-	6	
	7					7	
Right	8					8	Right

Straight Cable



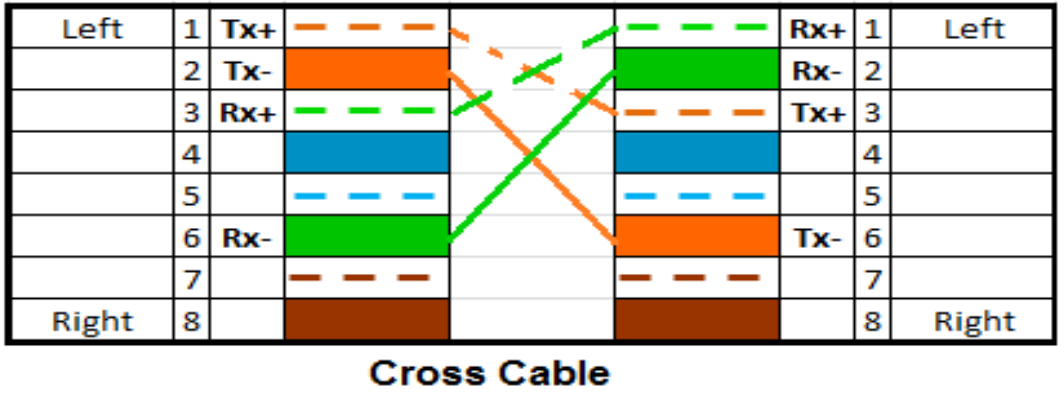
Crimping CAT cable

Cross Cable

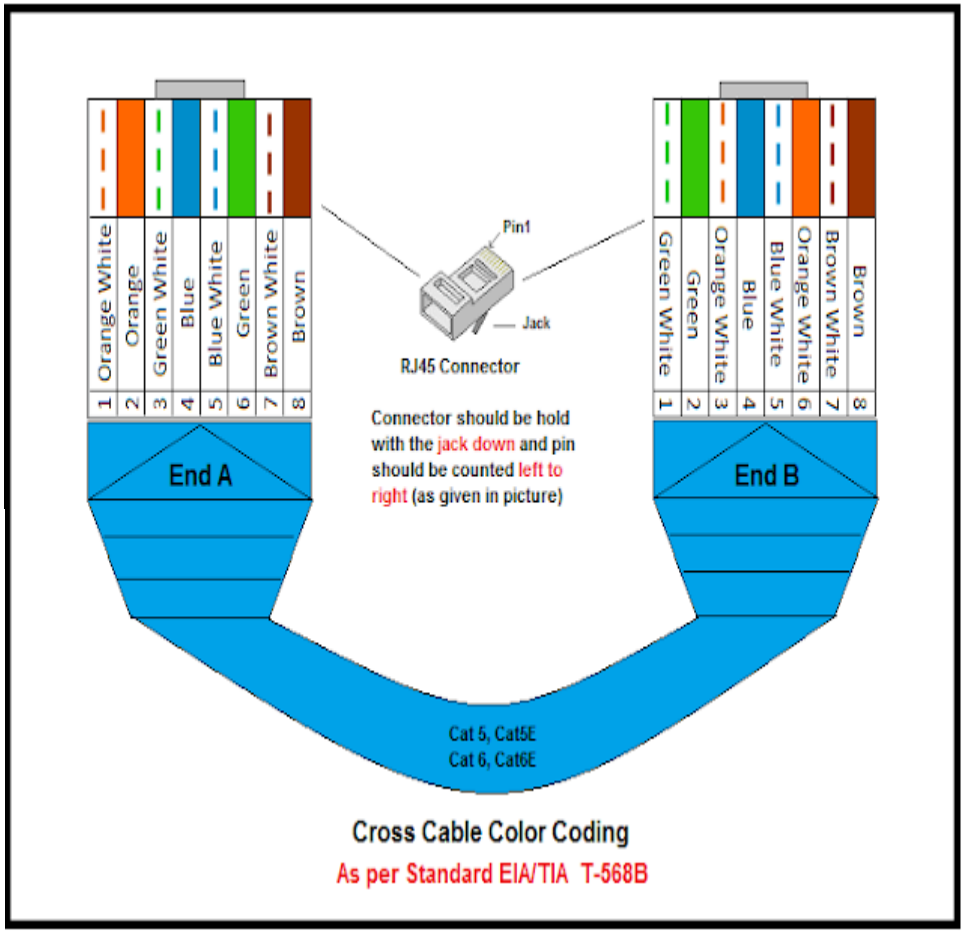
https://www.youtube.com/watch?v=LXcbZ_l5E0U

Used to connect devices with same functions or roles

- Connecting uplinks between switches.
- Connecting PC to PC
- Connecting PC to a Router
- Connecting 2 routers together without a a

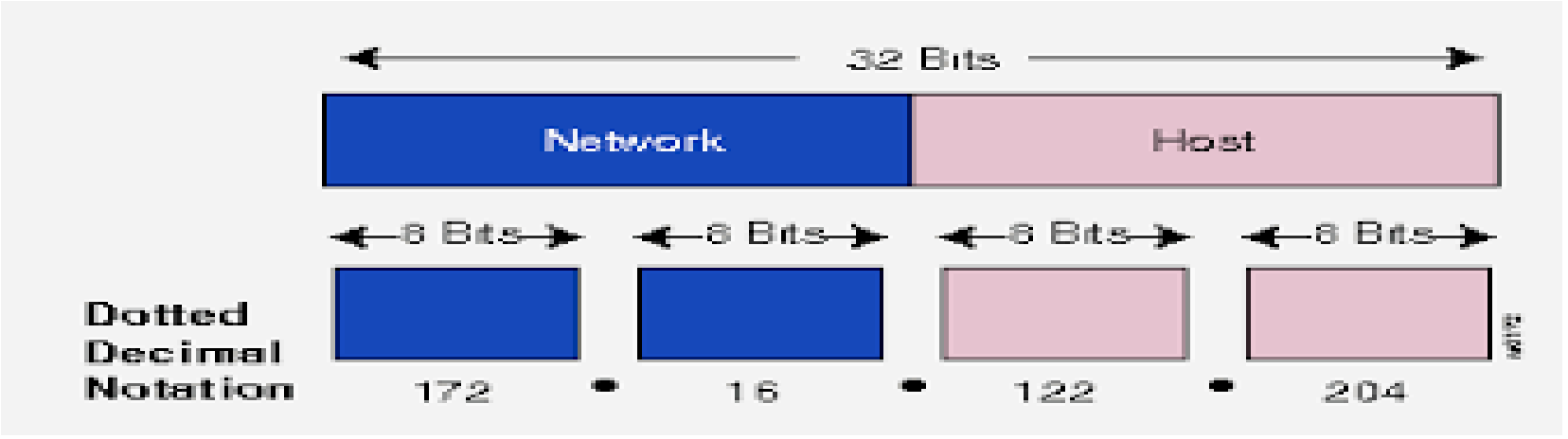


Cross Cable



Study of IP Address

IP Address Format



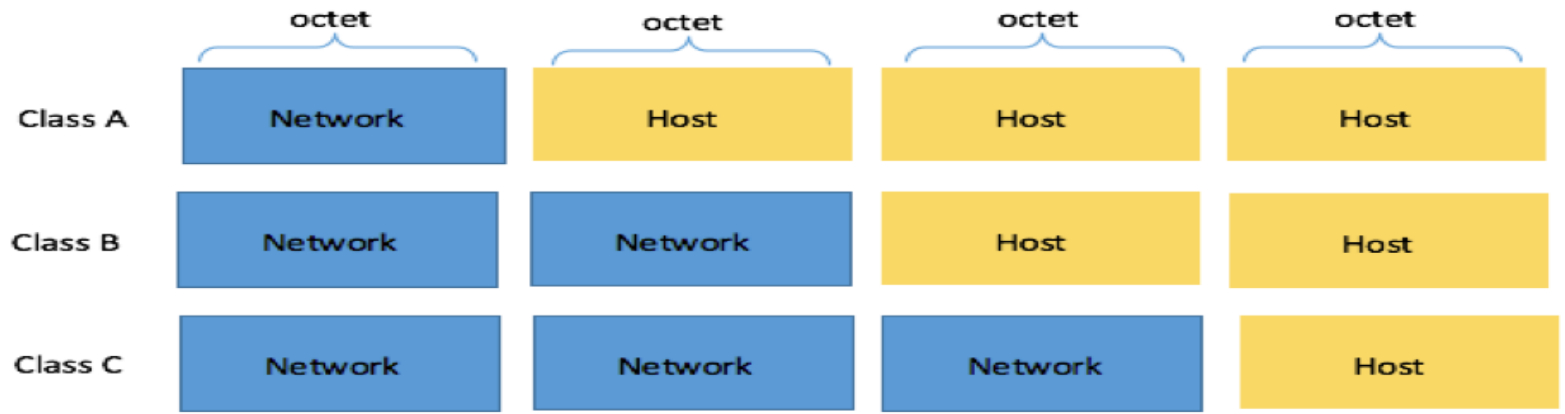
The **Network ID** part of IP address identifies the physical network to which the computer is attached

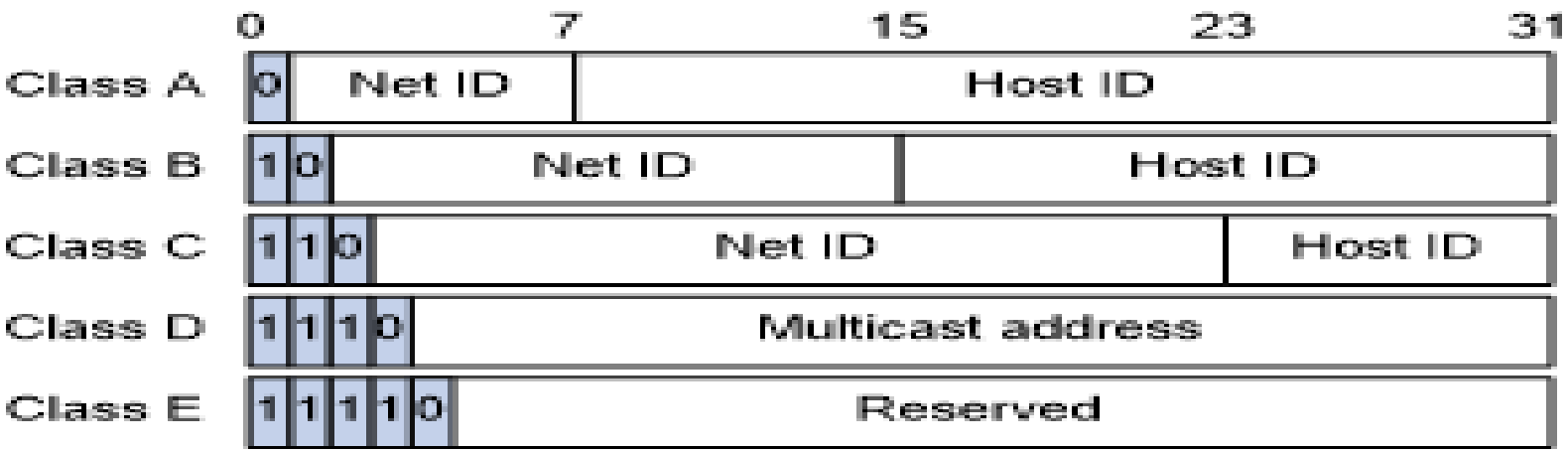
The **Host ID** part identifies the individual computer on the network

Three classes of IP address are used practically CLASS A, B and CLASS C

Study of IP Address

IP4 Address Class





Study of IP Address

IP4 Address Class

Class	1st octet of IP address	Default Subnet Mask	Network / Host	Number of networks	Maximum nodes in a network
A	1 - 126	255.0.0.0	N.H.H.H	126	16,777,214
B	128 - 191	255.255.0.0	N.N.H.H	16,384	65,534
C	192 - 223	255.255.255.0	N.N.N.H	2,097,152	254
D	224 - 239				
E	240 - 254				

0 & 127 are not used in class A , because starting with 127 has its different kinds of uses

Non-Routable IPs used in Local Area Network

Class	Private IP address range	Subnet mask
A	10.0.0.0 – 10.255.255.255	255.0.0.0
B	172.16.0.0 – 172.16.31.255	255.255.0.0
C	192.168.0.0 – 192.168.255.255	255.255.255.0

Study of IP Address

Subnet Mask

- The subnet mask splits the IP address into the Network ID and Host ID
- Routers in the network extracts the Network ID part using this subnet mask and forwards the packets to the designated network
- A subnet mask is a 32-bit number created by setting host bits to all 0s and setting network bits to all 1s.
- A Class A subnet mask reflects the network portion in the 1st octet and host portion in octets 2, 3, and 4 and so on

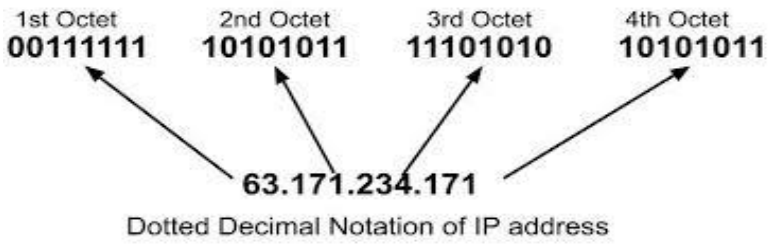
Class A: 11111111.00000000.00000000.00000000 → 255.0.0.0

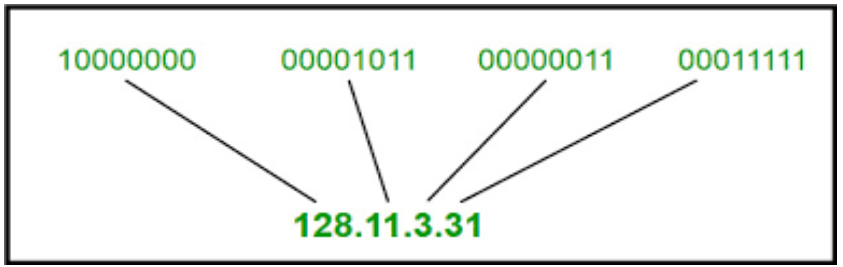
Class B: 11111111.11111111.00000000.00000000 →1255.255.0.0

Class C: 11111111.11111111.11111111.00000000 →255.255.255.0

Study of IP Address

IP4 Address Examples



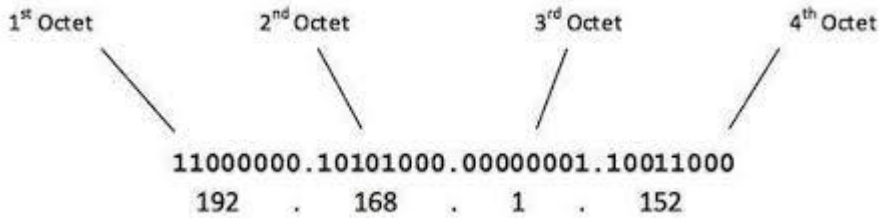


Class A address

Subnet Mask 255.0.0.0

Class B address

Subnet Mask 255.255.0.0



Class C address

Subnet Mask 255.255.255.0

Study of IP Address

Subnetting

- It is the technique for logically partitioning a single physical network into multiple smaller sub-networks or subnets.

Advantages :

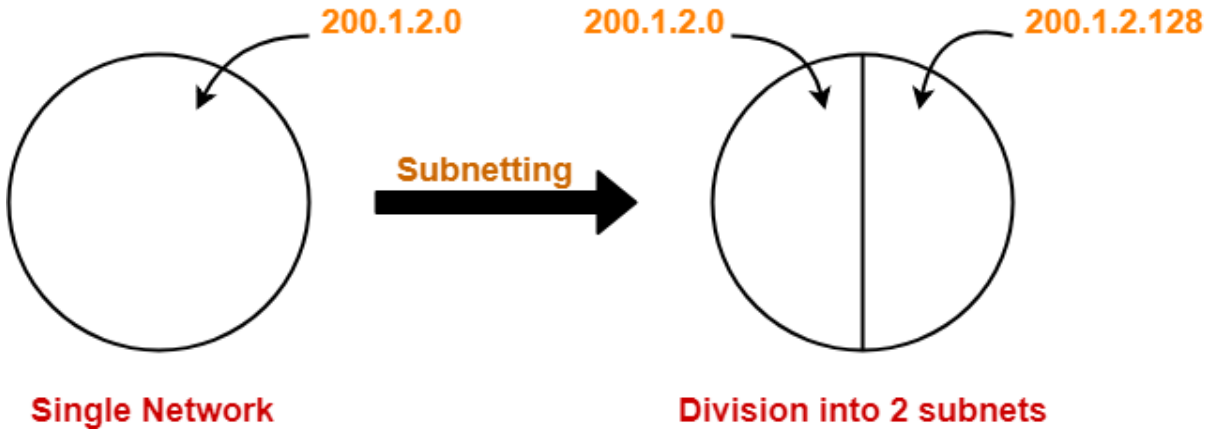
- Networks, maintenance is easy
- Reducing broadcast volume and thus network traffic
- It protects accessing one network from another network . For example, MCA Dept. N/W can not be accessed at CS Dept. N/W

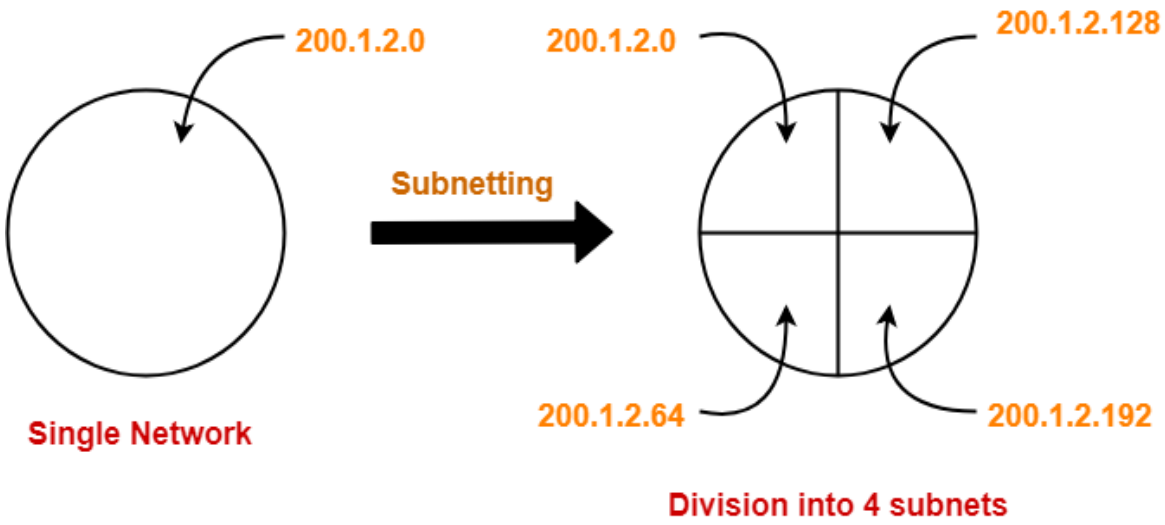
https://www.subnet-calculator.com/

Study of IP Address

Sub-netting

.63
127
191
255





Study of IP Address

For 1st Subnet

IP Address of the subnet
= 200.1.2.0
Total number of IP Addresses
= 2^7 = 128
Total number of hosts that can be configured
= 128 – 2 = 126
Range of IP Addresses
= 200.1.2.1 to 200.1.2.126
Broadcast Address
= 200.1.2.127
Subnet Mask
= 255.255.255.128 OR /25

For 2nd Subnet-

IP Address of the subnet
= 200.1.2.128
Total number of IP Addresses
= 2^7 = 128
Total number of hosts that can be configured
= 128 – 2 = 126
Range of IP Addresses
= 200.1.2.129 to 200.1.2.254
Direct Broadcast Address
= 200.1.2.255
Subnet Mask
= 255.255.255.128 OR /25

Study of IP Address

Subnet-1
N/W-Id=200.1.2.0
Host / subnet = 2^6-2 = 64-2=62
Host Ips = 200.1.2.1 to 200.1.2.62
Broadcast IP = 200.1.2.63

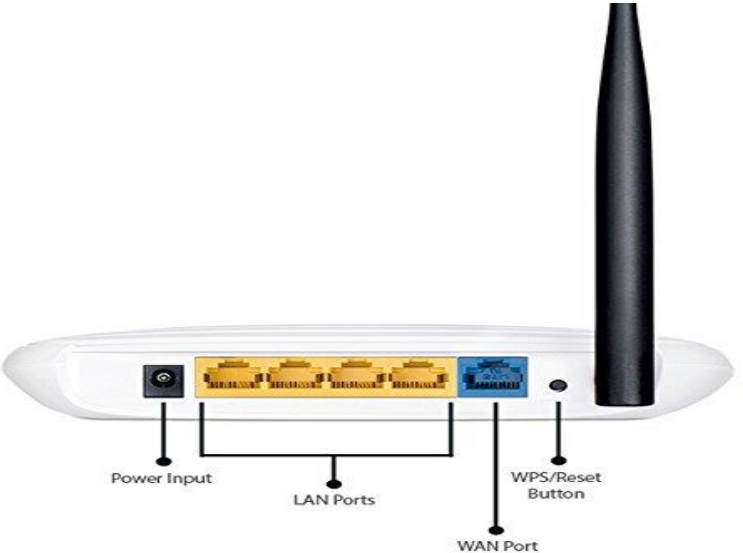
Subnet-2
N/W-Id=200.1.2.64
Host / subnet = 2^6-2 = 64-2=62
Host Ips = 200.1.2.65 to 200.1.2.126
Broadcast IP = 200.1.2.127

Subnet-3
N/W-Id=200.1.2.128
Host / subnet = 2^6-2 = 64-2=62
Host Ips = 200.1.2.129 to 200.1.2.190
Broadcast IP = 200.1.2.191

Subnet-4
N/W-Id=200.1.2.192
Host / subnet = 2^6-2 = 64-2=62
Host Ips = 200.1.2.193 to 200.1.2.254
Broadcast IP = 200.1.2.255

Wireless LAN



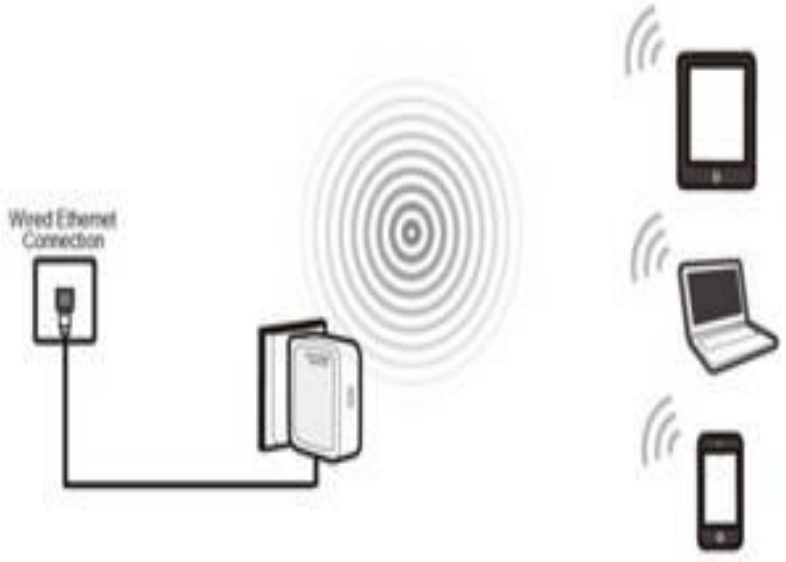






Wireless LAN

TP-Link devices can operate in different modes



AP (Access Point) Mode for extending the Network

- AP mode is more used to transfer wired connection into wireless.
- It works like a **switch**. Usually, it is behind a router.

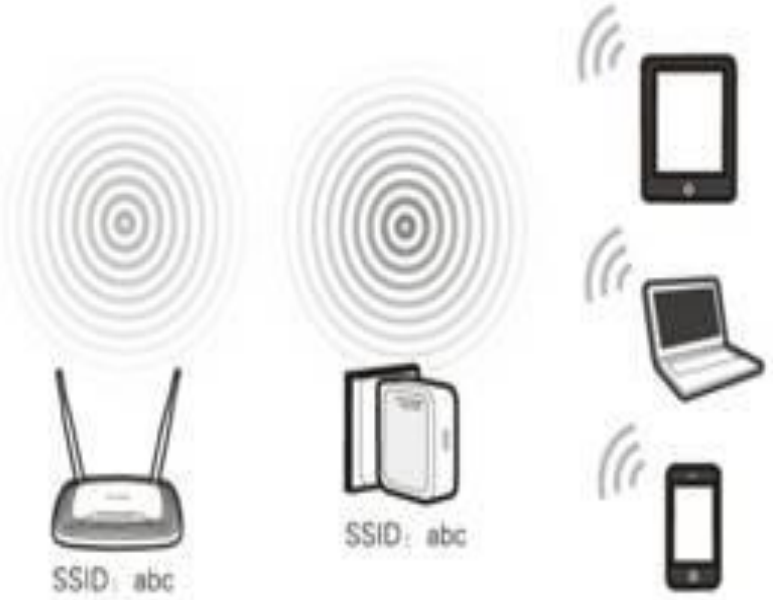
Wireless LAN



Wireless Router Mode(for home Internet sharing)

- With the router mode, it can share one wired Internet connection to several clients.
- Connect Wired internet point to WAN port.

Wireless LAN

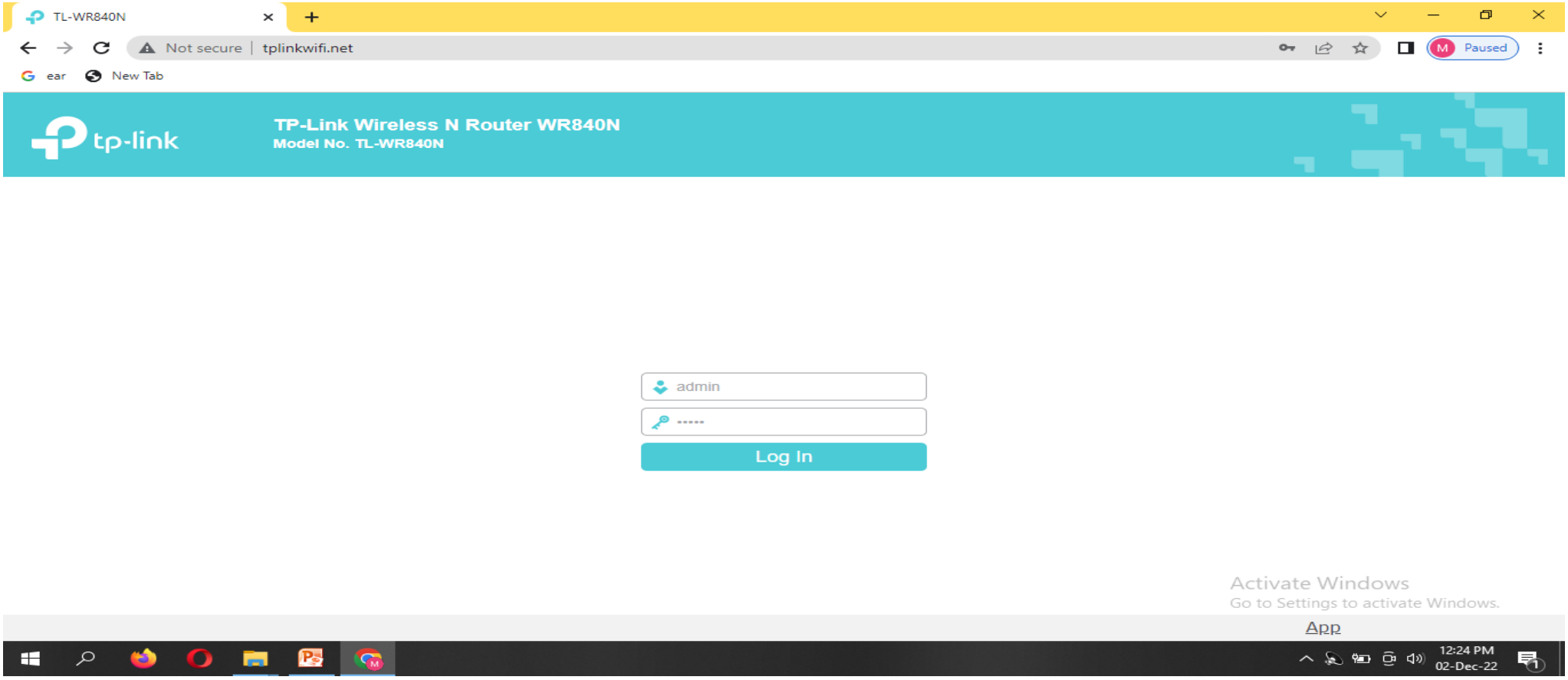


Repeater mode/ Range Extender
(for home Wi-Fi extension)

Repeater mode is used to extender the wireless coverage with same SSID and security.

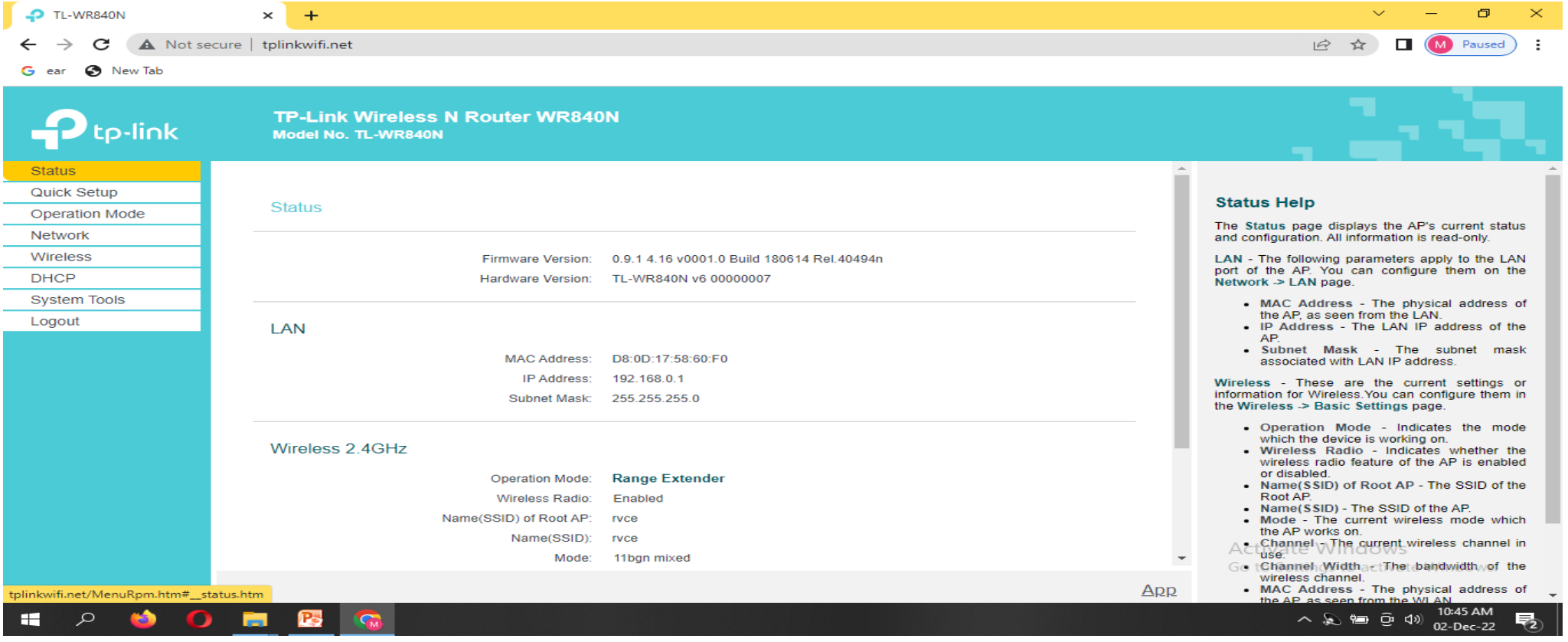
Wireless LAN

In Browser- http//tplinkwifi.net
Login with admin and admin



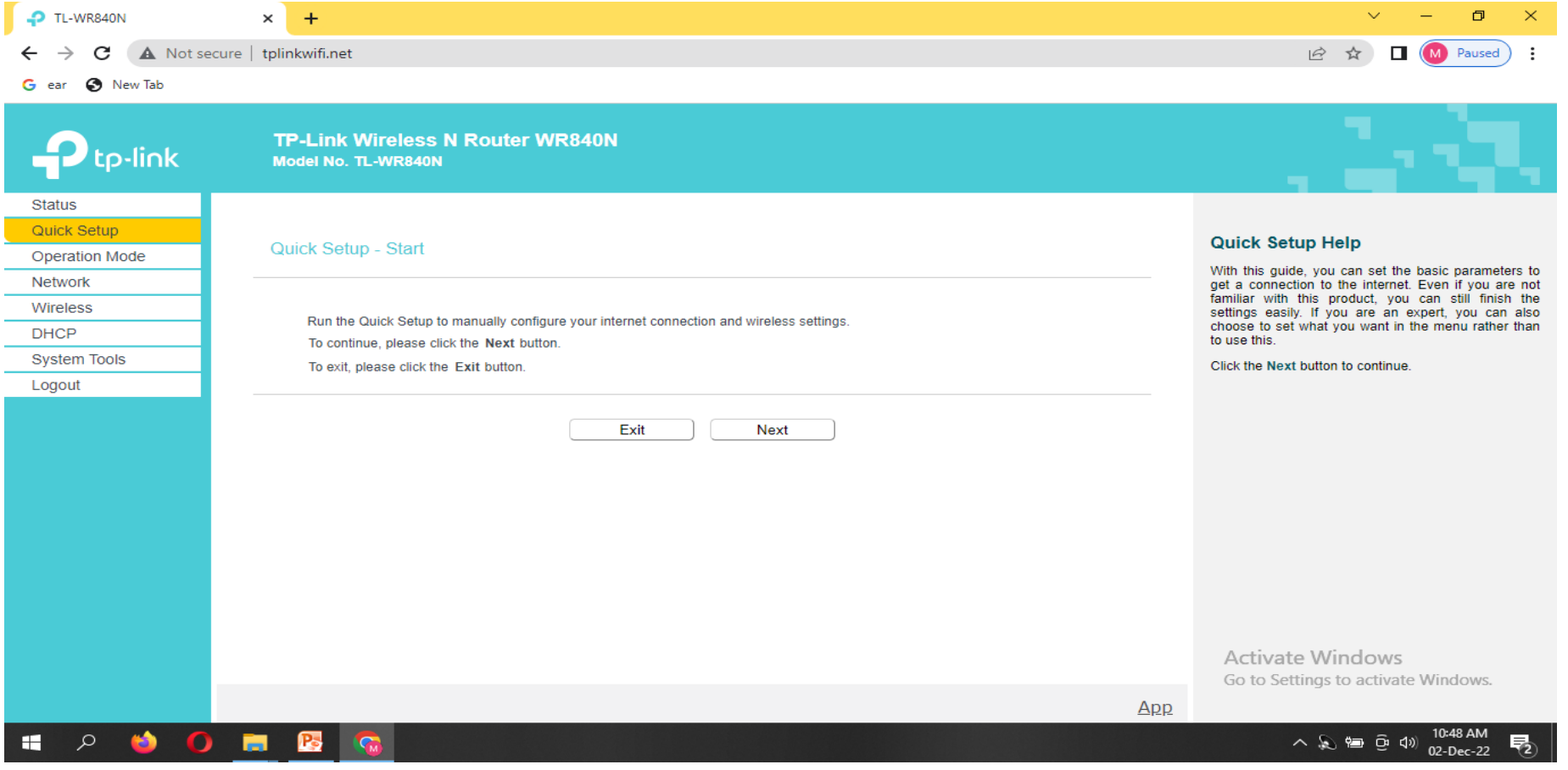
Wireless LAN

Current status of the Device



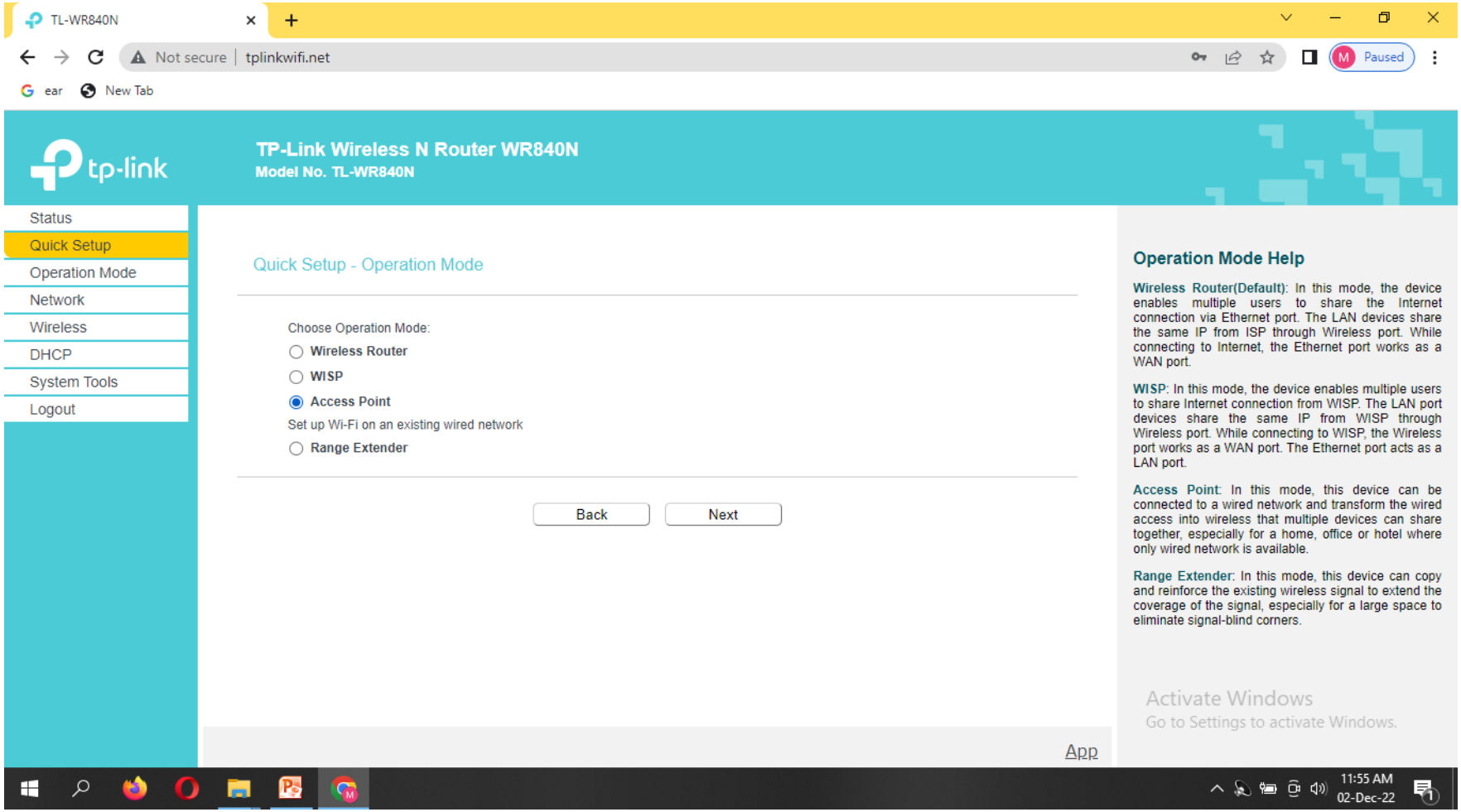
Wireless LAN

Choose Quick Setup



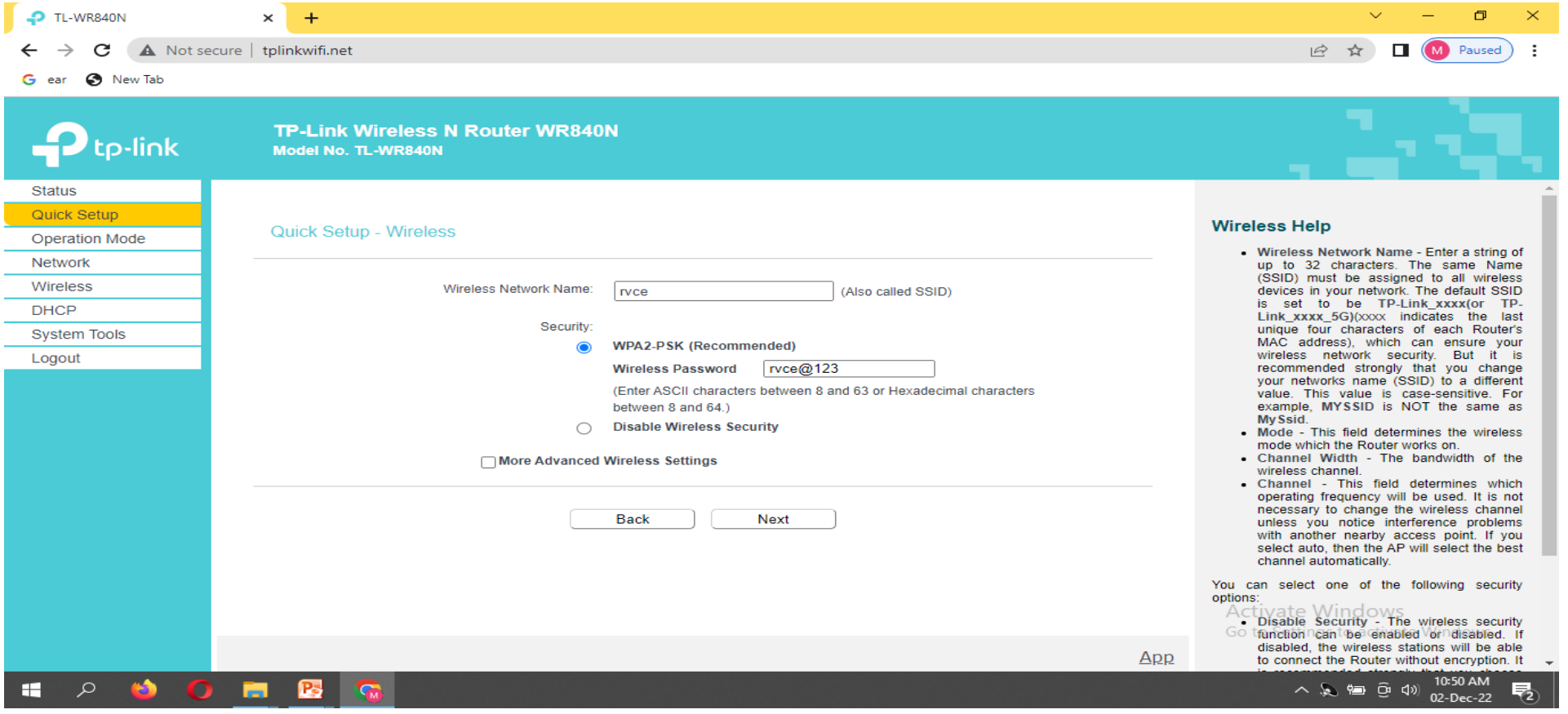
Wireless LAN

Set mode as Access Point



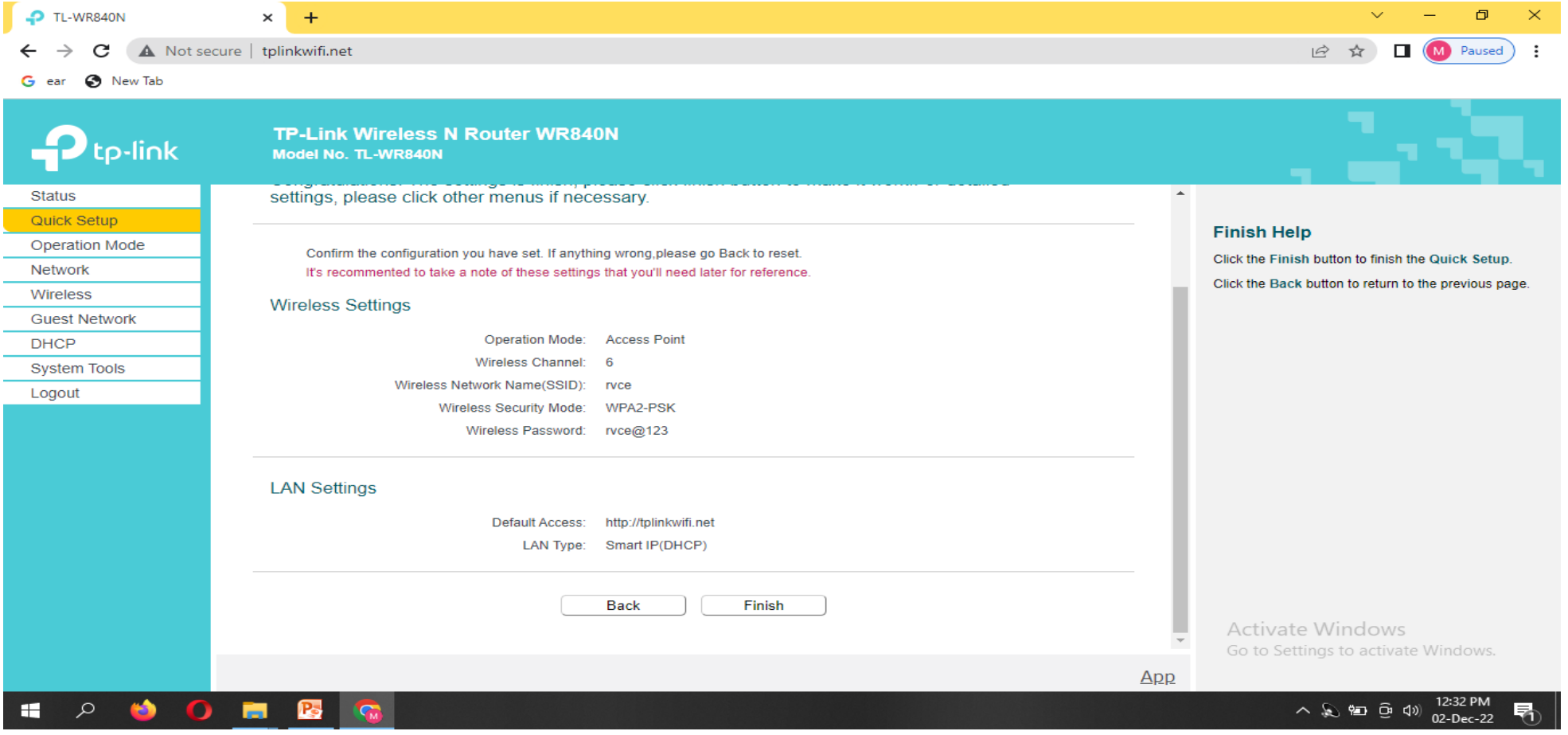
Wireless LAN

Change Service Set Identifier (SSID)/ **Wi-Fi network name & Set password**

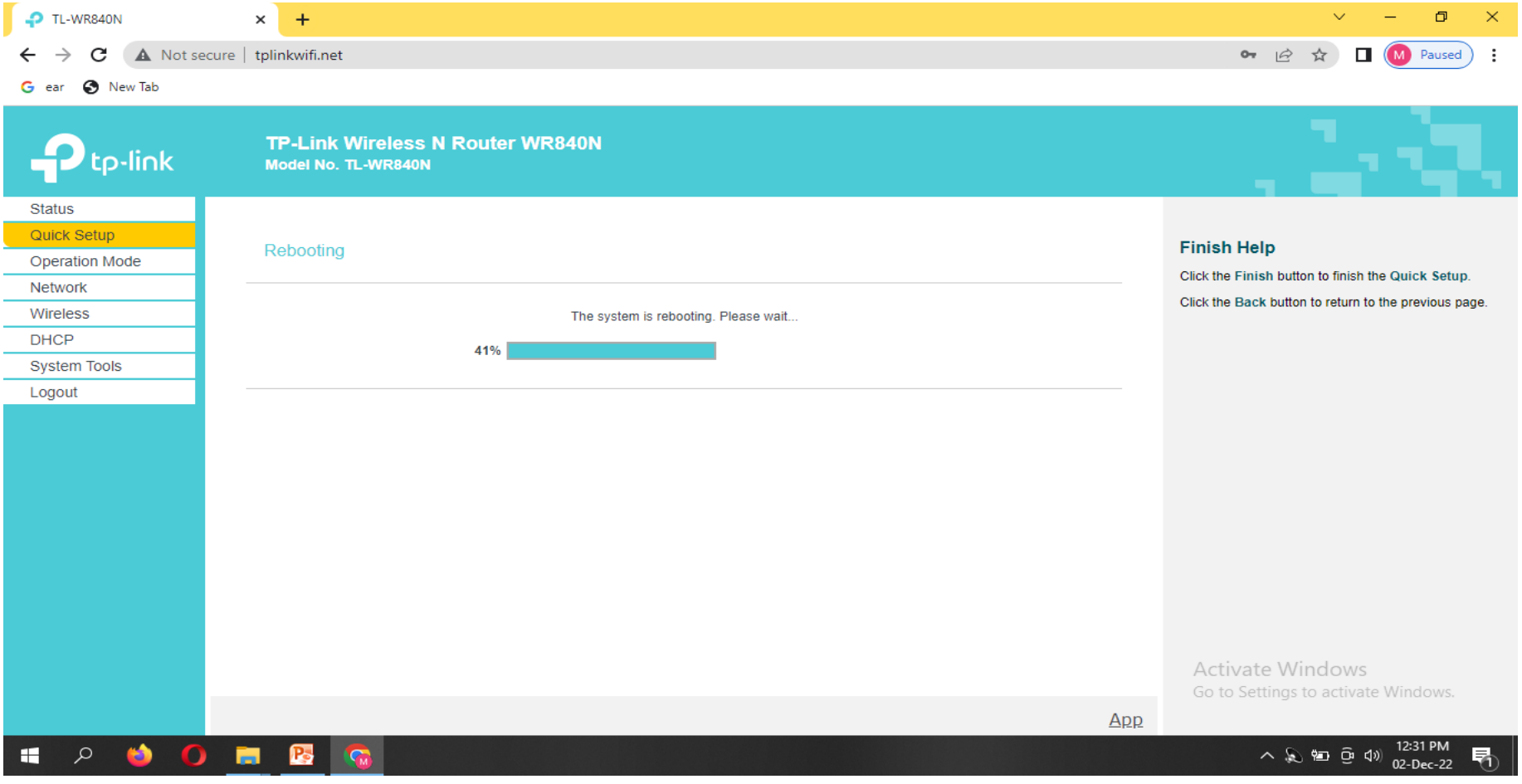


Wireless LAN

Preview the settings and click on Finish



Wireless LAN



Wireless LAN

Now AP is ready to build WLAN

